State-of-Health Estimation of Lithium-Ion Batteries Using Incremental Capacity Analysis Based on Voltage–Capacity Model

Jiangtao He, Zhongbao Wei, *Member, IEEE*, Xiaolei Bian, and Fengjun Yan, *Member, IEEE*

***Abstract*— State of health (SOH) is critical to evaluate the life expectancy of lithium-ion battery (LIB), thus should be estimated accurately in practical applications. This article proposes a computationally efficient model-based method for SOH estimation of LIB. A revised Lorentzian function-based voltage–capacity (VC) (RL-VC) model is exploited to accurately capture the voltage plateaus of LIB which reflect the material-level phase transition phenomenon. A full set of new features of interest (FOIs) is extracted by simply fitting the RL-VC model leveraging data collected from the constant-current charging process. Correlation analysis is then performed for the captured FOIs, based on which linear models are calibrated to estimate the battery SOH. The proposed method is validated with experimental data from different battery chemistries. The results show that the extracted FOIs have high linearities with the battery capacity, suggesting a good potential for SOH estimation and better feasibility over traditionally used methods. The proposed method shows a high accuracy for battery SOH estimation and an expected robust performance against the initial aging status and practical cycling condition.**

***Index Terms*— Feature extraction, incremental capacity analysis (ICA), lithium-ion battery (LIB), state of health (SOH), voltage–capacity (VC) model.**

I. Introduction

CHARACTERIZED by the high volumetric and gravimetric energy densities, lithium-ion batteries (LIBs) have been at the forefront of applications for the end-user sector electrification such as electrical vehicles (EVs). State of health (SOH) is a critical indicator in the battery management system (BMS) of LIB system. The accurate estimation of SOH provides users with valuable advice about the necessary maintenance or replacement [1].

The capacity has been viewed as a suitable indicator for the SOH considering its direct link to the millage of EVs. However, the SOH is difficult to be estimated in an *in situ* manner. Model-based estimators have been extensively studied in the literature and have shown expected performance [2]–[5]. However, such architectures are sensitive to the modeling accuracy, which cannot be guaranteed under dynamic road missions and over the life cycle of battery. In contrast, data-based methods, such as the Gaussian process regression (GPR), neural network (NN), and useful features of interest (FOIs) extraction to infer the SOH, could release from such concerns attributed to the model-free property [6]–[10]. For these machine learning algorithms such as GPR and NN, the main challenge is the reliability and practicability since the accuracy of these estimators largely depends on the data training process. The complex driving conditions in EVs will make it difficult and time consuming to ensure the estimator's accuracy. Alternatively, the FOIs extracted from the data during the standard constant-current–constant-voltage (CCCV) charging process have been used for SOH estimation with no data training process needed. Eddahech *et al.* [11] used a simple exponential function to fit the CV charging curves. Wang *et al.* [12] utilized the equivalent circuit model (ECM) to extract useful FOIs from the CV phase and further to estimate the SOH. Yang *et al.* [13] used the ECM to derive the current time constant, which was further used to indicate the SOH. However, the CV phase occupies a large proportion of the whole charging time, thus the acquisition of complete CV data is difficult.

Comparatively, incremental capacity analysis (ICA) and differential voltage analysis (DVA) based on the CC charging data are proposing candidates among data-based methods for SOH estimation. With respect to DVA-based methods [14]–[19], peaks and valleys in DV curves are less identifiable, and the breakpoints for the capacity-related area are hard to determine. Furthermore, the DV curve refers to the capacity which keeps fading over time [20]. To this end, ICA methods which transform voltage plateaus into observable peaks are more commonly used in practical applications. In particular, the peaks and valleys of IC curves were leveraged to analyze multiple aging mechanisms, e.g., the loss of lithium inventory (LLI) and the loss of active material (LAM) [21]–[24]. It was also shown that the amplitude and position of IC peaks mitigated regularly over the lifetime of $LiFePO_4$ (LFP) battery [25], [26]. This motivates the use of IC features to estimate the SOH by either mapping FOIs to the capacity directly [27]–[29] or using data fusion methods such as sparse Bayesian model [30] and GPR [31]–[33].

It is noted that the ICA method faces obvious barriers in real applications, i.e., the raw IC curves typically contain a

Manuscript received December 30, 2019; revised February 29, 2020 and April 19, 2020; accepted May 8, 2020. Date of publication May 14, 2020; date of current version June 19, 2020. *(Corresponding authors: Zhongbao Wei; Fengjun Yan.)*

Jiangtao He and Fengjun Yan are with the Department of Mechanical Engineering, McMaster University, Hamilton, ON L8S 4L8, Canada (e-mail: yanfeng@mcmaster.ca).

Zhongbao Wei is with the National Engineering Laboratory for Electric Vehicles, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: weizb@bit.edu.cn).

Xiaolei Bian is with the KTH Royal Institute of Technology, 11428 Stockholm, Sweden (e-mail: xiaoleib@kth.se).

Digital Object Identifier 10.1109/TTE.2020.2994543

large amount of noises due to the derivative operation. This fact solicits increased interests and endeavors to come up with filtering techniques for data pretreatment. To date, a variety of prefiltering methods have been proposed to smooth the IC curves, such as Butterworth filter [34], moving average (MA) [27], [31], and Gaussian filter [27], [31]. The curve smoothness, however, is sensitive to the sampling interval and the parameters of the filter. There lacks mathematical guarantee for the fulfillment of curve smoothness and feature retention, which are both critical to the FOI extraction. The explicit difference existed in the aging data sets further hinders the exploitation of a universal set of filtering parameters that are applicable to the entire life of LIB. Moreover, the FOI extraction after direct ICA is both tedious and subjective, which is not unfavorable for accurate SOH estimation.

To remedy aforementioned deficiencies, voltage–capacity (VC) models have been raised to estimate the SOH based on smoothed IC curves. Weng *et al.* [26], [35] used the support vector machine (SVM) to get the smoothed IC curves with moderate computing effort and further proposed a unified open circuit voltage (OCV) model to capture the aging features. Guo *et al.* [36] parameterized the ECM based on the CC charging data, while one of the identified parameters was used to estimate the SOH. Torai *et al.* [25] presented a model based on the deformed pseudo-Voigt peak function to describe the IC curve of LFP. Feng *et al.* [37] utilized the probability density function (pdf) to evaluate the SOH, and an equivalence was found between the PDF and ICA methods. Such methods give deep insights into the data-based SOH estimation, although they are relatively complicated from the real-time point of view. A simple yet accurate VC model was proposed by Li *et al.* [38], [39], and its revised version was presented by Bian *et al.* [40]. Such VC models are initially aimed for fitting the IC peaks and offer much scope for the aging analysis of LIB. The potential of them for multitype FOI extraction and SOH estimate is yet entirely out of consideration in state of the art.

This article aims to bridge aforementioned knowledge gaps and proposes a practical method for data-based SOH estimation of LIB. The revised Lorentzian function-based VC (RL-VC) model is exploited to capture the informative FOIs hidden behind the CC charging data and further to estimate the SOH of LIB. Compared with the work of Li *et al.* [38], [39], the Lorentzian function is revised in this article to improve the robustness of the VC model against the uncertain boundary setting. More importantly, while the works from [38] and [39] focus on capacity modeling, this article emphasizes on the RL-VC model-based extraction of a new set of informative FOIs, correlation analysis, and the use of FOIs for accurate SOH estimation. Two primary contributions have been made in this article. First, a full set of new FOIs is extracted by leveraging the RL-VC model based on onboard collected currents and voltages during CC charging. Second, multitype FOIs are analyzed in terms of their linearity with the reference capacity and the potential for SOH estimation. The newly proposed FOIs are validated to be more efficient than the traditional ones in terms of SOH estimation. To the best of the authors' knowledge, this is the first attempt of using the VC model to capture multitype FOIs and further to estimate the SOH.

The remainder of this article is organized as follows. Section II introduces the RL-VC model and the method for SOH estimation. Section III briefs the experimental data set used for method development and validation. The results of FOI analysis and SOH estimation are given in Sections IV and V, respectively, while the major conclusions are drawn in Section VI.

II. Methodology

A. Incremental Capacity Analysis

The potential of LIB electrode depends on the thermodynamic property of electrode material, thus the potential range and voltage plateaus are unique for different electrode materials at different life stages. While the electrode potentials are difficult to measure, the OCV which represents the differential potential between cathode and anode contains valuable information about the material-level phase transition over the life cycle of LIB. Hence, the OCV curve, if well evacuated, is insightful to reflect the SOH of LIB. However, the direct use of OCV for health diagnostic is challenging. First, the intercalating and deintercalating of lithium ions happen only over a narrow voltage range. Second, all voltage plateaus lie in and overlap within the critical voltage ranges, while the voltage is vulnerable to measurement noises. All these factors lead to a quasi-observable feature of the OCV curve of most LIB chemistries.

ICA is commonly used to extract the implicit information from the easy-to-obtain terminal voltage curve. The original IC curve, i.e., $V \sim dQ/dV$, can be obtained by applying the derivative operation to the measured Q–V curve. From the mathematical point of view, the derivative should be approximated by the difference operation given by [27]

$$\frac{dQ}{dV}\Big|_k \approx \frac{\Delta Q_k}{\Delta V_k} = \frac{Q_k - Q_{k-1}}{V_k - V_{k-1}} \tag{1}$$

where Q is the instantaneous charged capacity, and V is the terminal voltage of the battery.

Based on the above transmission, multiple peaks can be found in the IC curve, while each peak represents a particular voltage plateau of LIB. The peak features are associated directly with the phase transition of electrode material and presents high sensitivity to the aging of battery. Upon observing the changes of peak, the aging origins such as LAM and LLI can be analyzed, and then the SOH can be estimated.

B. RL-VC Model

The IC curve determined directly by (1) is subject to large amount of noises due to the poor numerical condition caused by the derivative operation at $\Delta V \approx 0$. The noises can be attenuated by either enlarging the sampling interval or filtering the original curve. However, the involved parameters have to be tuned meticulously, and the optimality of them cannot be guaranteed over the entire life of LIB. Moreover, the extraction of FOIs needs extensive postprocessing. For example,

TABLE I
PARAMETERS INVOLVED IN THE LORENTZIAN FUNCTION AND THEIR IMPLICATIONS

Parameter	Description
$\boldsymbol{n}$	number of peaks, determined by the phase transition dynamics
$\boldsymbol{A_i}$	area of i-th peak, describing the accumulated charge during i-th phase transition of active materials
$\boldsymbol{V_{0i}}$	symmetry center of i-th peak, indicating the redox potential of phase transition
$\boldsymbol{\omega_i}$	full width at half maximum of the corresponding transition processes

the extraction of peak area brings lots of uncertainties, since the starting and ending points for integral are highly subjective.

In seeking to remedy aforementioned barriers, this article aims to reshape the IC curve and to extract informative FOIs for SOH estimation in a model-based framework. The morphological feature of IC curves of commonly used LIBs suggests the potential of using suitable peak fitting functions to mimic the IC curves. The Lorentzian function can be employed to fit the IC curve, that is,

$$\frac{dQ}{dV} = \sum_{i=1}^{n} \frac{2A_i}{\pi} \frac{\omega_i}{\omega_i^2 + (2V - 2V_{0i})^2} \tag{2}$$

where the description of involved parameters is summarized in Table I.

It is noted that only the current and terminal voltage are directly measurable in real applications. To facilitate direct calibration thereby, integral is performed to (2), giving the Lorentzian-VC (L-VC) model:

$$Q = \sum_{i=1}^{n} \frac{A_i}{\pi} \arctan\left(\frac{2(V - V_{0i})}{\omega_i}\right) + C \tag{3}$$

where C is a constant as a consequence of the integral. On this premise, an adjustment to (3) within the same model framework gives the RL-VC model as follows [40]:

$$Q = Q_{\max}\left[\sum_{i=1}^{n} \frac{a_i}{\pi} \arctan\left(\frac{2(V - V_{0i})}{\omega_i}\right) + C_r\right] \tag{4}$$

where $Q_{\max}$ is the CC charge capacity-related value, while a_i denotes the weight of each peak component. In spite of the similar expression, the RL-VC model can be expected to enhance the error tolerance against the parameter boundary.

C. Parameter Identification and Correlation Analysis

The parameters involved in (4) have to be identified to achieve expected fitting accuracy. Generally, n is determined by the material attributes of cathode and anode in the battery. The LFP cathode is known to have a two-phase feature with one flat voltage plateau, while the graphite anode has four voltage plateaus. As a result, n should be four to cover all phase transition features of LFP battery [25], [36]. However, the optimal peak number of the RL-VC model should be determined meticulously for the purpose of SOH estimation. One reason is that some of the peaks are too weak to be observable. Apart from this, there may be either peak vanishment or new peak emergence during the aging of battery. For the illustrative purpose, the raw and smoothed IC curve based on the MA filter [27], [31] is presented in Fig. 1(a). Furthermore, the smoothed IC curves under different aging cycles are plot in Fig. 1(b), where three main peaks are labeled as Peak I, Peak II, and Peak III. As the aging cycle reaches 493, a subpeak named as Peak I_1 emerges associated with Peak I. However, from the SOH estimation point of view, the emerged subpeak is less valuable for the aging detection due to the low observability and behaves more likely a disturbance. In this regard, n is selected as three for the LFP battery in this article. In the case of Li(NiCoAl)O2 (NCA) battery, only two significant peaks can be found on the IC curves [41], the n value is set to two.

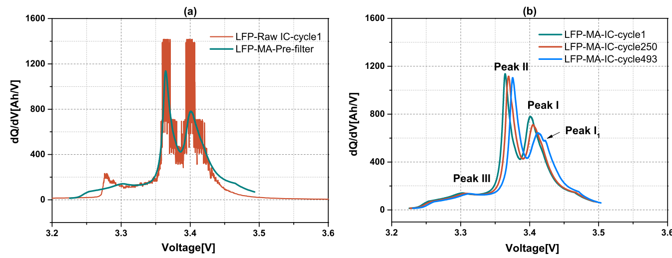

Fig. 1. IC curves obtained from LFP battery. (a) Raw IC versus smoothed IC. (b) MA smoothed IC curves for different aging cycles.

The above discussions give recommended selections of the peak number for the modeling of LFP and NCA batteries. A smaller peak number will cause the failure of capturing all useful FOIs. In contrast, any excessive selection may cause the inefficient capture of less-insightful FOIs under the best situation, while under the worst situation, it potentially causes the over-fitting problem which eventually results in the loss of meaning of extracted FOIs.

The nonlinear least squares (NLSs) algorithm is used for parameter identification. The boundaries of parameters for LFP battery and NCA battery are listed in (5) and (6), respectively. Here, Q_{end} means the measured charged capacity in the CC charging process. In this article, the upper and lower bounds of fitting parameters are determined empirically in (5) and (6), and an expected performance can be observed for the entire life cycles of both LFP and NCA batteries. Hence, the nonlinear fitting is not that sensitive to the boundary setting. Generally, a precalibration of the model can suggest a reasonable boundary condition that ensures the fitting performance over long cycling cycles:

$$\text{LFPbattery}\begin{cases} a_i \in [0, 1] \\ V_{0i} \in [3, 3.65] \\ \omega_i \in [0, 0.2] \\ C_r \in [0.5, 0.52] \\ Q_{\max} \in [1, 1.1] * Q_{\text{end}} \end{cases} \tag{5}$$

$$\text{NCAbattery}\begin{cases} a_i \in [0, 1] \\ V_{0i} \in [3, 4.2] \\ \omega_i \in [0, 0.2] \\ C_r \in [0.5, 0.52] \\ Q_{\max} \in [1, 1.1] * Q_{\text{end}} \end{cases} \tag{6}$$

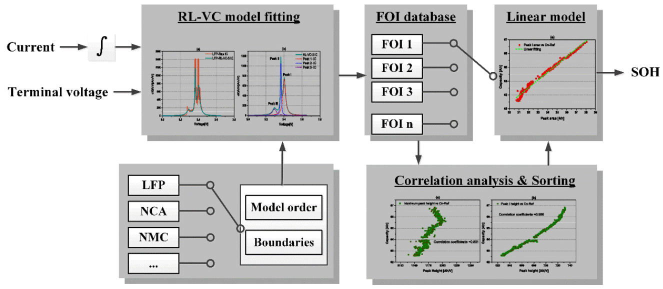

Fig. 2. Overall flowchart of the proposed SOH estimation method.

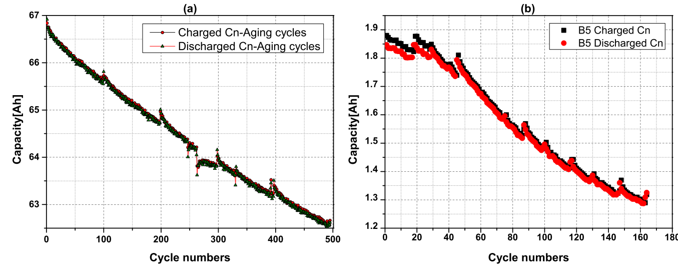

Fig. 3. Battery aging data sets. (a) LFP charge/discharge aging capacity. (b) NCA-B5 charge/discharge aging capacity.

TABLE II
CORRELATION COEFFICIENTS BETWEEN FOIS AND THE REFERENCE CAPACITY FOR NCA-B5 DATA SET

FOI	Correlation coefficient
maximum peak height	0.989
peak I height	0.984
peak II height	0.985
maximum peak voltage	-0.985
peak I voltage	-0.965
peak II voltage	-0.985
peak I area	0.982
peak II area	0.992

Upon obtaining the FOIs from the RL-VC model, the degree of the linearity between a specific FOI and the reference capacity, also called as the Pearson product–moment correlation coefficient, can be calculated by

$$R_i = \frac{\sum_{i=1}^{n}\left(\mathrm{FOI}_i - F\bar{O}I\right)\left(C_{n,i} - \bar{C_n}\right)}{\sqrt{\sum_{i=1}^{n}\left(\mathrm{FOI}_i - F\bar{O}I\right)^2}\sqrt{\sum_{i=1}^{n}\left(C_{n,i} - \bar{C_n}\right)^2}} \tag{7}$$

where the range of R_i is $[-1, 1]$. A correlation coefficient close to $+1/-1$ indicates a strong positive/negative correlation, while a value close to 0 indicates the loss of correlation.

D. Architecture of the Proposed Method

The proposed method can be generalized into the following steps, which are shown schematically in Fig. 2.

Step 1: The model order and parameter boundaries given by (5) and (6) are first determined based on the chemistry of the investigated battery.

Step 2: The resultant RL-VC model is used to fit the raw V–Q data, which is obtained from the current and terminal voltage measurements during the CC charging process. A full set of FOIs is extracted via the model parameterization, as shown in Table II.

Step 3: The correlation analysis and FOI sorting are performed based on the FOI-SOH data sets at different aging levels, to find the optimal FOI for SOH estimation. A linear model is afterward trained to fit the relationship between the optimal FOI and the reference SOH.

Step 4: In practical applications, the optimal FOI is extracted from the onboard measurements, and the SOH is estimated by inputting the captured FOIs to the calibrated linear model obtained from Step 3.

III. EXPERIMENTAL DATA SET

The proposed method is validated by experimental data from both LFP battery and NCA battery. The first data set [40] is obtained by the China Electric Power Research Institute on an LFP battery, whose nominal capacity and voltage are 60 Ah and 3.2 V, respectively. The aging tests were repeated for 500 times using 0.5 C for charge/discharge within the voltage bound of 2.5–3.65 V, and the data sampling frequency is 0.2 Hz. The dynamic temperature environment is set to mimic the actual working conditions and to accelerate the battery degradation, the details can be found in [40]. The maximum charge/discharge capacities (C_{n_chg} and C_{n_disch}) obtained from the aging tests are plotted in Fig. 3(a), where it is explicit that the C_{n_chg} curve is almost overlapped with the C_{n_disch} curve. The slight deviation, if observable, is explained by the fact that the columbic efficiency can never be 100%.

The second data set is from NASA Prognostics Center of Excellence (PCoE), where three NCA batteries (B5, B7, and B18), whose nominal capacity and voltage are 2 Ah and 3.7 V were cycled through three different operational profiles at room temperature, respectively [42]. All the three cells were charged under CCCV mode, where the CC current, upper voltage, and terminal current are 1.5 A, 4.2 V, and 20 mA, respectively. In contrast, the CC discharge was applied at 2 A until the terminal voltage drops to 2.7, 2.2, and 2.5 V for cells B5, B7, and B18, respectively. It is noted that the first charged capacities of the three cells are 1.88, 1.92, and 1.873 Ah, indicating an inconsistency from the beginning stage, respectively. The reference charged/discharged capacities of B5 as a representative are plotted in Fig. 3(b).

Upon obtaining the aging data, the proposed method can be implemented easily. With respect to the algorithmic procedures summarized in Section II-D, Steps 1–3 consist of offline battery testing, model calibration, correlation analysis, and linear fitting. The NLS-based parameter identification is performed by calling the lsqnonlin function in MATLAB. The correlation analysis is performed by (7), while the linear model is calibrated by the commonly used batch least squares method. Step 4 consists of an FOI extraction process and a simple linear model-based prediction, which can be easily executed with the commonly used embedded microcontroller units.

IV. RESULTS OF FOI EXTRACTION AND ANALYSIS

This section focuses on the model validation, FOI extraction, and the potential of different FOIs for estimating the SOH.

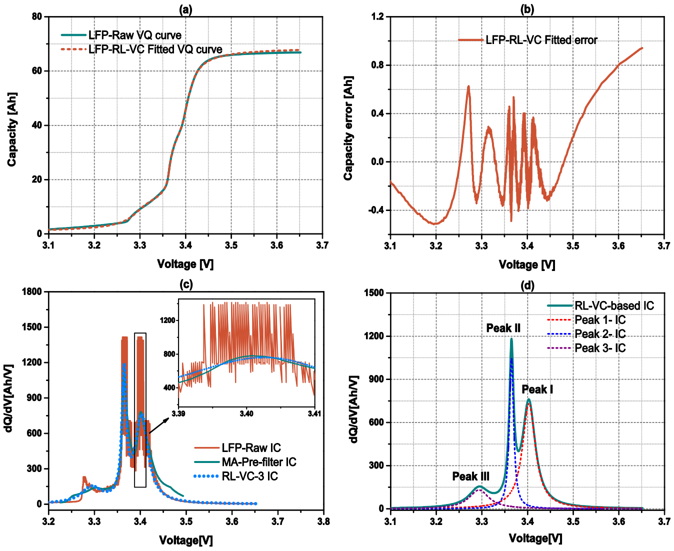

Fig. 4. IC curves obtained from LFP battery. (a) RL-VC model fit $V{-}Q$ curve. (b) $V{-}Q$ fitting error. (c) IC curves based on three different methods. (d) Decomposed peak components.

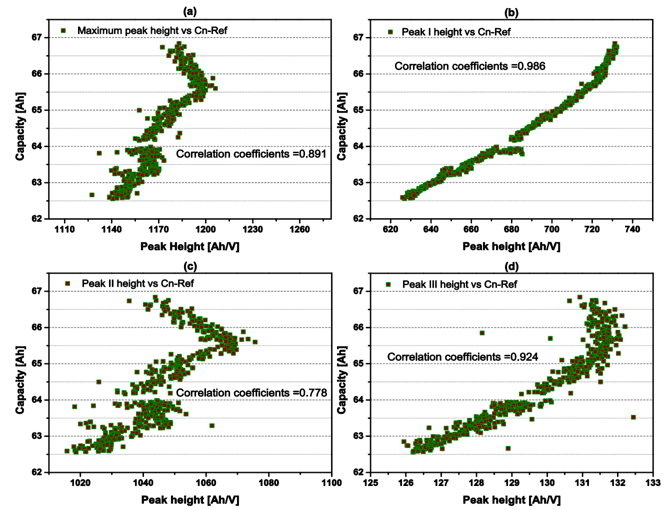

Fig. 5. Relationship of peak heights and the reference capacity of LFP battery. (a) Maximum peak height versus C_n. (b) Peak I height versus C_n. (c) Peak II height versus C_n. (d) Peak III height versus C_n.

A. VC Model Validation and FOI Extraction

A full set of FOIs can be obtained directly by parameterizing the RL-VC model leveraging the raw data collected from the CC charging phase. The reliable FOI extraction, however, lays the foundation on an accurate modeling of IC curve which contains fruitful information about the phase transition of active electrode materials during the aging process. For illustration purposes, the RL-VC model-fit $V{-}Q$ curve of LFP battery in comparison with its measurement is shown in Fig. 4(a), while the fitting error is shown in Fig. 4(b). It is shown that the $V{-}Q$ curve has been modeled accurately with the error well confined to 0.92 Ah error bound, corresponding to a relative error of 1.53%. The raw IC curve, as well as the ones given by the prefiltering method and the RL-VC model-based method, is plotted comparatively in Fig. 4(c). It seems that both methods have introduced large deformations at peaks I and II, as the peak magnitude of raw IC curve seems much larger than the two smoothed curves. It should be noted that this is a visual pseudomorphism. As can be observed from the zoomed-in figure of peak I in Fig. 4(c), the seemly high peak magnitude of raw IC curve is due to the extremely large noises caused by both the voltage measurement error and the quasi-ill numerical condition of $\Delta V \approx 0$. The large fluctuations and spikes are actually disturbances that cannot reflect any aging mechanism. To this end, the proposed method attenuates the disturbance by adopting an LS-based fitting. The generated IC curve, albeit seems discounting the peak magnitude, can better reflect the strength of electrode material phase change more reasonably.

It should be noted that the extracted FOIs are different from those in traditional ICA-based methods. Traditional ICA-based methods can only extract FOIs from the overlapped peaks. In contrast, the RL-VC model-based method decomposes the raw charging data directly into three peak components, as shown in Fig. 4(d). The full set of FOIs corresponding to different peak components, including the peak heights, peak positions, and peak areas, can then be determined simultaneously by simply parameterizing the RL-VC model. Hence, the RL-VC model-based method gives rise to a new set of FOIs different from existing studies. As each separate peak in the IC curve is directly tied to the material phase change at a particular voltage plateau, the newly extracted FOIs are theoretically more efficient than previously used ones in the literature. This is also validated from the estimation results in this article.

B. FOI Analysis for LFP Battery

The relationship between extracted peak heights and reference capacities for LFP battery is plotted in Fig. 5. The peaks I–III are obtained from the RL-VC model fitting, while the traditionally used maximum peak height is found from the smoothed IC curve. It is observed that the peak I height has the highest linearity with the reference capacity, suggested by the high correlation coefficient of 0.986. In comparison, relatively weak linearities can be found in Fig. 5(a), (c), and (d). Interestingly, the commonly used maximum peak height is not optimal since the correlation coefficient is only 0.891. The results challenge the entrenched mindset of using directly observable FOIs to estimate the SOH, which is widely adopted by the existing ICA methods. Also, it is worth to mention that C_n means the CCCV charged capacity of the cycled data.

The relationship between extracted peak voltages and reference capacities for LFP battery is plotted in Fig. 6. It is shown that the absolute correlation coefficients are larger than 0.98 for all these cases, suggesting the high linearities of peak voltages with the reference capacity. Notably, the maximum peak voltage and peak II voltage are preferable to indicate the SOH considering the high absolute correlation coefficient of 0.992. The relationship between extracted peak areas and reference capacities for the LFP battery is plotted in Fig. 7. It is explicit that peak I area has strong linearity with the reference

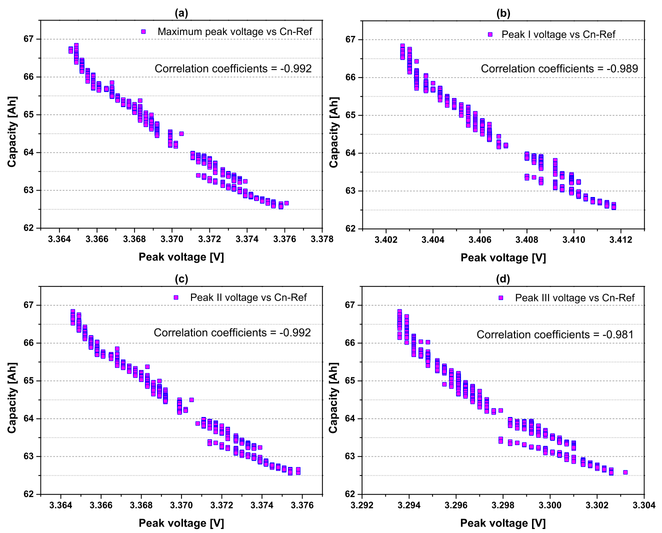

Fig. 6. Relationship of peak voltages and the reference capacity of LFP battery. (a) Maximum peak voltage versus C_n. (b) Peak I voltage versus C_n. (c) Peak II voltage versus C_n. (d) Peak III voltage versus C_n.

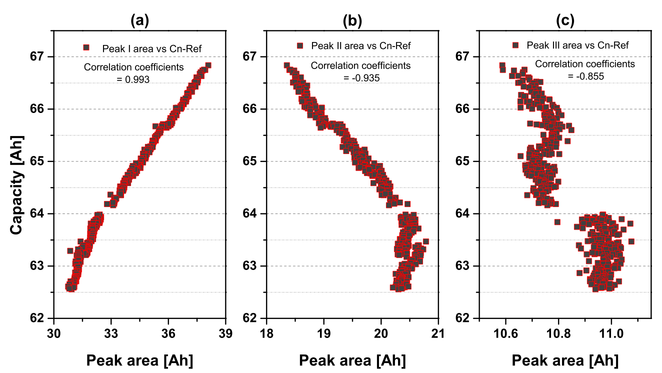

Fig. 7. Relationship of peak areas and the reference capacity of LFP battery. (a) Peak I area versus C_n. (b) Peak II area versus C_n. (c) Peak III area versus C_n.

capacity. In comparison, the evolution of peaks II and III areas lacks a monotony with respect to the reference capacity, thus their potential to indicate the SOH is less promising. The correlation analysis suggests that the proposed method can capture several new FOIs, which are highly informative to reflect the change in the aging status.

C. FOI Analysis for NCA Battery

In terms of the NCA battery, the NCA-B5 aging data set is used for extracting and analyzing the FOIs based on the RL-VC model. The correlation coefficients between multiple FOIs and the reference capacity are summarized in Table II. Unlike the case of LFP battery, most of the extracted FOIs have high linearity with the reference capacity. In each category of FOIs, the maximum peak height, peak II voltage, and peak II area are preferable considering their higher correlation coefficients over other candidates in the same category. The drift of the IC curve alongside the aging is plotted successively in Fig. 8(a). The IC peaks drift toward higher voltage along with the aging, which can be explained by the increased

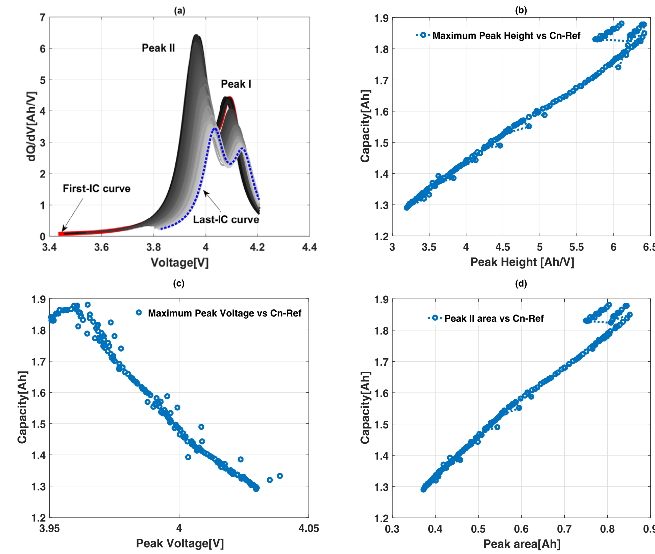

Fig. 8. FOI evolution along with the aging of NCA battery. (a) IC curves. (b) Maximum peak height versus C_n. (c) Peak II voltage versus C_n. (d) Peak II area versus C_n.

internal resistance due to the LLI and LAM. The high linearity of selected FOIs, i.e., the maximum peak height, peak II voltage, and peak II area, with the reference capacity, can be reflected straightforwardly by observing Fig. 8(b)–(d).

V. Results of SOH Estimation

This section focuses on the SOH estimation based on selected FOIs in Section IV. The robust performance of the proposed method against the change in the initial aging status and working condition is also discussed.

A. SOH Estimation Results for LFP Cell

A linear model can be established to map the selected FOI to the SOH of the battery. In this case, the SOH can be estimated efficiently by inputting the extracted FOIs in real applications to the calibrated linear model. In the case of LFP battery, peak I height, peak II voltage, and peak I area are selected as significant FOIs to illustrate their implication for SOH estimation. The corresponding estimation results are shown in Fig. 9. It is observed that the estimated SOH values closely resemble the available measurements for all the investigated cases. The mean absolute errors (MAEs) of reference SOH estimation are 0.226%, 0.163%, and 0.159%, by using peak I height, peak II voltage, and peak I area as the FOI, respectively. The maximum errors have been well confined to 1.1% error bound for all the cases. Moreover, the highest accuracy can be achieved by using peak I area as the FOI, with an MAE of 0.159%. This is rooted in the highest linear correlation coefficients of 0.993, as suggested in Fig. 7(a). However, one may want to adjust the priority of FOI according to the practical condition, since all the selected FOIs have demonstrated high fidelity for the SOH estimation. The estimation results suggest that the proposed method is quite promising for real applications from the engineering

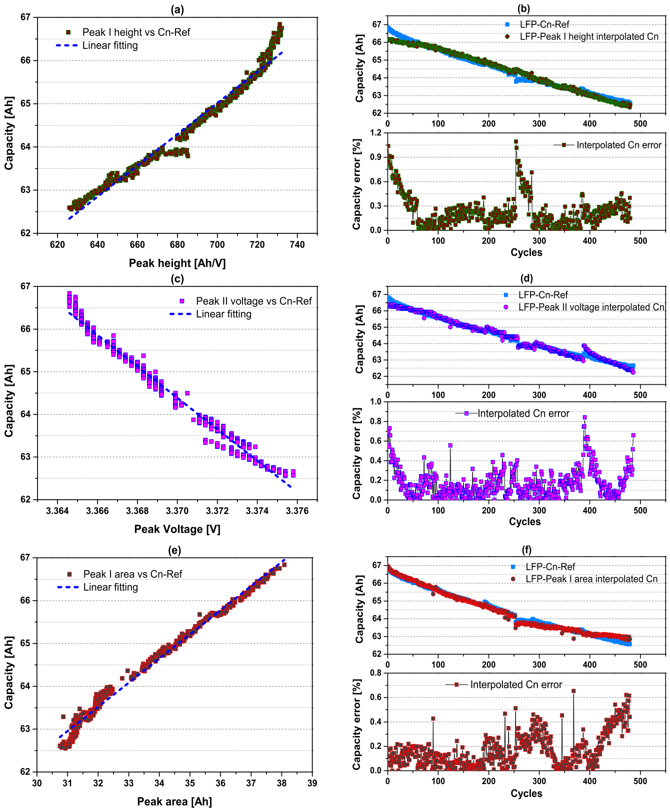

Fig. 9. SOH estimation results for LFP battery. (a) Peak I height linear model. (b) Estimation results based on peak I height. (c) Peak II voltage linear model. (d) Estimation results based on peak II voltage. (e) Peak I area linear model. (f) Estimation results based on peak I area.

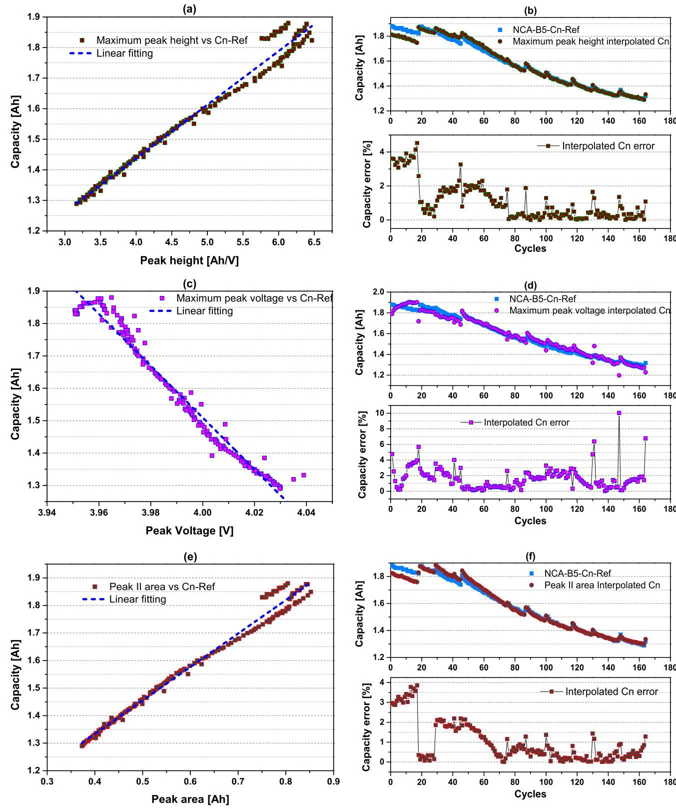

Fig. 10. SOH estimation results for NCA battery (B5). (a) Maximum peak height linear model. (b) Estimation results based on maximum peak height. (c) Peak II voltage linear model. (d) Estimation results based on peak II voltage. (e) Peak II area linear model. (f) Estimation results based on peak II area.

viewpoint, considering the sufficiently high estimation accuracy and the foreseeable low computing cost of linear model.

B. SOH Estimation Results for NCA Cell

In the case of NCA battery, the maximum peak height, peak II voltage, and peak II area are selected to estimate the SOH, considering their high linearities with the reference capacity in each FOI category. The estimation results are shown in Fig. 10. It is observed that all curves are quasi-linear except for some points obtained at a high SOH level. Such irregular points are attributed to the unsuitable setting of the parameter boundaries, which causes uncertain parameterization. This problem, albeit occurs only at initial cycles, can be avoided by empirically adjusting the boundaries of critical parameters.

Several important conclusions can be made in terms of the SOH estimation results. First, the reference capacities of NCA cells have been tracked accurately over the entire life span by using the proposed method. The MAEs of reference SOH estimation are 0.336%, 0.470%, and 0.309% by using maximum peak height, peak II voltage, and peak II area as the FOI, respectively. Second, the highest accuracy can be achieved by using peak II area as the FOI. This is rooted in the highest linear correlation coefficients of 0.992. Considering the high linearities as suggested in Table II, the FOIs can be selected with high freedom with the awareness that a high correlation coefficient generally gives a high estimation accuracy of SOH. Consistent results are obtained by using the proposed method on the B7 and B18 data sets. However, details are not repeated here for simplicity.

C. Robustness Analysis

It is known that cells with the same chemistry may deviate from each other rooted in the manufactural process. Such cell inconsistency can be enlarged remarkably along with the aging process. In this regard, it is highly imperative to evaluate the proposed method on the robustness to the cell inconsistency. Particularly, the linear model obtained from B5 is used directly to estimate the SOH of cell B7 and B18. It is noted that the initial aging status and the cycling conditions are all different for the three cells, as readily explained in Section III.

The relationship between peak II area and the reference capacity obtained from different NCA cells is shown in Fig. 11. It is observed that the linearity has been well kept in spite of the change in the initial aging status and cycling condition. Moreover, the evolution of peak II area versus C_n from different cells follow the consistent path, but reasonable deviations are still observable. The observed variations may be attributed to the random selection of cells from different batches. The consistency is expected to be higher in practical applications since the cells are typically screened before the packing process to ensure sufficient consistency. The SOH estimation results based on the two new cells are shown

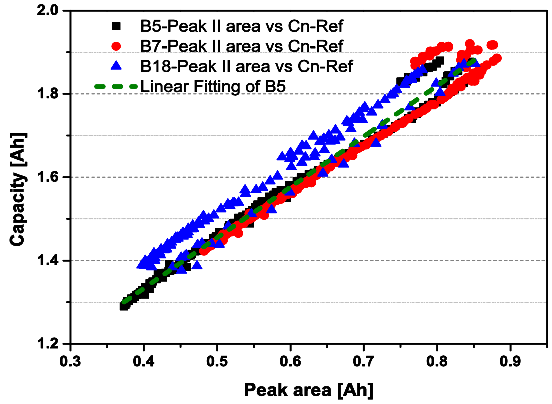

Fig. 11. Relationship between peak II area and reference capacity obtained from different cells.

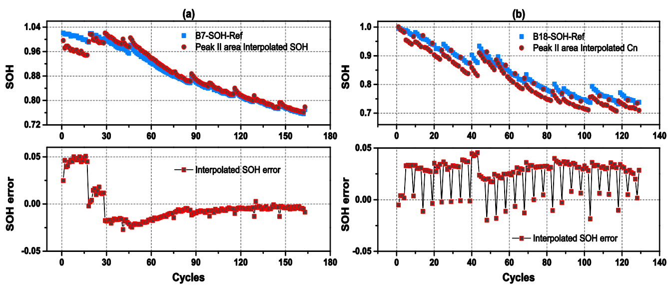

Fig. 12. SOH estimation results based on peak II area for NCA battery. (a) B7. (b) B18.

TABLE III
MAE OF IC CURVE FITTING AND C_n ESTIMATION BASED ON DIFFERENT VC MODELS

Model	Model fitting MAE(Ah)		SOH MAE (%)	
	LFP	NCA	LFP	NCA
RL-VC model	0.197	0.011	0.159	0.309
L-VC model	0.287	0.017	0.324	1.103

in Fig. 12. As shown, the developed linear model from cell B5 can be well generalized to the other cells, even though they are different in the initial aging status and cycling condition. The MAE of estimation is 1.348% and 2.577%. The robustness and generality of the proposed method are, hence, validated.

D. Comparison With Existing State of the Art

Compared with the work of Li *et al.* [38], [39], the Lorentzian function has been revised in this article to improve the robustness of the VC model against the uncertain boundary setting. The superiority of the RL-VC model over the original L-VC model can be validated by observing the accuracy of both the IC curve fitting and the SOH estimation, as summarized in Table III. It is explicit that the proposed RL-VC model outperforms the reported L-VC model with respect to the accuracy of both IC curve fitting and SOH estimation. The practical use of the proposed method is, hence, more promising.

The proposed VC model-based method outperforms the traditional prefiltering-based methods in terms of the practical feasibility. First, unlike the prefiltering methods whose parameters are highly subjective and nonoptimal during the life cycle of battery, the proposed method can ensure the optimality under the criterion of squared error minimization. Second, the prefiltering-based ICA needs extensive postprocessing on the filtered IC curve, and the determination of important peak areas is not practical considering the difficulty to locate the starting/ending points of integral. In contrast, the RL-VC model-based method is free from such barriers and decomposes the raw charging data directly into three peak components, as shown in Fig. 4(d). The full set of FOIs, including the peak heights, peak positions, and peak areas, can be determined simultaneously by simply parameterizing the VC model. From the feasibility point of view, therefore, the proposed method is more efficient than the commonly used ICA methods.

E. Discussions and Future Work

The proposed method is feasible in practical applications, as the CCCV charging is a standard protocol for the LIB system. The charging current is 0.5 and 0.75 C for LFP and NCA batteries in this article, and a good estimation result can be observed for both cases. In practical applications, the CC charging current is typically no more than 0.5 C under the normal charging mode. As the IC curve under lower C-rate better reflects the FOIs relative to aging, the proposed method is foreseeably feasible in practical charging conditions.

It is noted that a model trained with particular aging cycles is only feasible to the corresponding aging paths. The CC discharge is used as a special case in this article to illustrate how the proposed method is implemented. However, the driving condition is more complicated in real EV applications, which will lead to a different aging path compared to the CC discharge. In practical applications of EVs, therefore, the aging database used for training the model should be well designed to cover similar aging paths as those in practical use, for example, using the typical driving conditions such as the Federal Urban Driving Schedule (FUDS) for discharge and the CCCV mode for charge. This is a feasible approach to address the uncertainty in driving conditions and has already been reported in the existing literature [28], [34]. In an extreme case that the future utilization scenario is totally unclear, the proposed method can be also extended to address the mentioned uncertainties. As reported by the existing literature such as [21], the SOH can be estimated by combining different FOIs of IC curves regardless of the aging paths. This is achieved by building a high-dimension database including selected FOIs, aging paths, and the resultant SOHs by computer simulations. The built database is then used directly in a BMS as a lookup table to fit the newly obtained FOIs. Such methods can well address the uncertainty in driving condition, and explicitly, their application depends on the accurate capture of informative FOIs which is one of the primary contributions of this article. Hence, in the most unexpected case, the proposed method is feasible by incorporating with the aforementioned database. This is an important topic from the engineering point of view and will be focused in our future work.

The temperature has an impact on the FOIs extracted from the IC curve fitting. For example, as temperature changes, the peak voltage of IC curve will mitigate due to a comprehensive effect of OCV rise and resistance drop. Therefore, there should be multiple FOI-SOH linear models under different temperatures. In other words, the temperature effect can be compensated by setting the coefficients of linear model as a function of the temperature. This is more likely an engineering problem that already reported in the literature [28], [32]. Since the primary contributions and focus of this endeavor lie in the model-based FOI extraction and correlation analysis, the temperature effect has not been considered in this article. However, the temperature compensation can be easily incorporated into the proposed method referring to the existing studies.

VI. Conclusion

In this article, a novel RL-VC model-based method is proposed for computationally efficient SOH estimation of LIB. A full set of new FOIs has been captured by simply parameterizing the RL-VC model. The linearities of different FOIs with battery capacity have been scrutinized, and linear models have been proposed to estimate the SOH directly from the capture FOIs. The proposed method is validated with experimental data from both the LFP battery and the NCA battery. The major conclusions are summarized as follows.

1) The RL-VC model is highly precise to capture the voltage plateaus during charging and the associated features of IC curve. The V–Q modeling has been well confined to 1.53% relative error bound.
2) The newly extracted FOIs turn out to be informative to the SOH estimation. Their extractions are also more efficient than the traditionally used ones from the viewpoint of both feasibility and estimation accuracy.
3) Six FOIs exhibit high linearities with the capacity of the LFP battery, with the correlation coefficient larger than 0.98. Particularly, peak I area shows the best prospect for SOH estimation, where the MAE of estimation is 0.159% over the battery life.
4) Seven FOIs have high linearities with the capacity of NCA battery, with the correlation coefficient larger than 0.98. Notably, peak II area shows the best prospect for SOH estimation, where the MAE of the SOH estimation is 0.309% over the battery life.
5) The proposed method has been validated with high robustness to the initial aging status and the practical cycling condition.

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Jiangtao He received the B.E. degree in materials physics from the Taiyuan University of Science and Technology, Taiyuan, China, in 2010 and the M.E. degree in materials science and engineering from the Beijing University of Technology, Beijing, China, in 2013. He is currently pursuing the Ph.D. degree in mechanical engineering with McMaster University, Hamilton, ON, Canada.

His current research interests include battery modeling and state estimation for electric vehicles.

Zhongbao Wei (Member, IEEE) received the B.Eng. and M.Sc. degrees in instrumental science and technology from Beihang University, Beijing, China, in 2010 and 2013, respectively, and the Ph.D. degree in power engineering from Nanyang Technological University (NTU), Singapore, in 2017.

He was a Research Fellow with the Energy Research Institute, NTU, from 2016 to 2018. He is currently a Professor in vehicle engineering with the National Engineering Laboratory for Electric Vehicles, School of Mechanical Engineering, Beijing Institute of Technology, Beijing, China. He has authored more than 40 peer-reviewed articles. His current research interests include modeling, identification, state estimation, diagnostic for battery systems, and energy management for hybrid energy systems.

Xiaolei Bian received the B.Eng. and M.Sc. degrees in mechanical engineering from Jilin University, Changchun, China, in 2014 and 2017. He is currently pursuing the Ph.D. degree in chemical engineering with the KTH Royal Institute of Technology, Stockholm, Sweden.

His current research interest includes modeling and state estimation of lithium-ion batteries (LIBs).

Fengjun Yan (Member, IEEE) received the B.Sc. degree from the Department of Control Science and Engineering, Harbin Institute of Technology, Harbin, China, in 2004, the M.Sc. degree from the Department of Automotive Engineering, Tsinghua University, Beijing, China, in 2006, and the Ph.D. degree from the Department of Mechanical and Aerospace Engineering, The Ohio State University, Columbus, OH, USA, in 2012.

From 2006 to 2007, he worked as an Engineer with FEV Inc., Dalian, China. He joined McMaster University, Hamilton, ON, Canada, as an Assistant Professor in 2012 and promoted as a tenured Associate Professor in 2017. His current research interests include dynamic system modeling, estimation and control, advanced vehicle powertrain control, vehicle electrifications, and intelligent vehicles.